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Seasonal variation in microplastic abundance, morphology, polymers, and adsorbed contaminants in wastewater treatment sludges and biosolids

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Abstract

This study investigated the seasonal impact on microplastic (MP) characteristics and concentration in the sludge streams (primary, secondary (waste activated) and digested) at a municipal wastewater treatment plant (WWTP) in Ontario. MPs were recovered after digestion and density separation of primary, secondary and digested sludge samples and were quantified and characterized (shape, colour, polymer type, and chemical contaminants) using microscopy and FTIR. The primary sludge had the highest number of MP particles among the three sludges, regardless of the season. The results show that sludge samples collected during the spring season had the highest MP concentrations (1.5–2.5 times higher), along with increased levels of tire and road wear particles (TRWPs), attributed to snowmelt and stormwater. Black and blue were the frequently recorded MP colours, with no discernible seasonal trends among different sludge types. Fibres remained the predominant MP type irrespective of the season and the sludge type. FTIR analysis of the 20–30% subset of MPs also revealed that PET was the most common polymer, with significant amounts of PA, PE, and PP recorded. There were no significant seasonal changes in the polymer type of the MPs. The spectral analysis further revealed the presence of a range of pharmaceutical drugs, industrial chemicals, and emerging contaminants on MPs, often obscuring the polymer identification while highlighting the role of MPs as carriers of environmental contaminants. Seasonal dynamics play a crucial role in assessing the emission of MPs into aquatic and terrestrial ecosystems. Understanding these fluctuations enables the development of more targeted and effective intervention strategies that are responsive to periods of heightened risk and that mitigate their environmental impacts.

Keywords Polymer identification, Wastewater treatment plant, Microfibres, FTIR analysis, Emerging contaminants



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1 Introduction

Wastewater treatment plants (WWTPs) are an important pathway for MP movement through the environment, acting both as a point of accumulation and emission [1]. MPs in WWTPs originate from diverse sources depending on the receiving sewer network, such as household, industrial and stormwater runoff [2–4]. Personal care products and microfibres (MFs) are released during laundry, forming a major contributor to the MP load of the WWTPs [5, 6]. The concentration, size, shape and type of MPs in WWTPs depend on several factors, such as demography, urbanization, and treatment techniques adopted [7–9].

Although WWTPs are not designed to remove MPs, the literature indicates that a removal efficiency of over 95% can be achieved for capturing MPs from the influent stream [10]. It is essential to acknowledge that MP removal through wastewater treatment processes results in their accumulation in the sludge streams, rather than their complete removal. The MPs are mainly captured in sedimentation tanks, and as a result, they are present in biosolids that may be later land-applied for beneficial use, potentially impacting soil ecology and the broader environment [11–13]. Additionally, the remaining 5% of MPs that escape treatment are mainly MFs and smaller MPs, which are discharged into surface waters [14, 15]. In short, MPs that pass through WWTPs can potentially contaminate both aquatic and terrestrial ecosystems [16, 17].

MP concentrations and removal are influenced by a wide range of factors, and the MP removal performance differs among treatment plants. This is largely attributed to the differences in treatment processes and the sewershed characteristics in WWTPs. However, studies have also shown that for the same treatment processes, the removal efficiency varies among treatment plants [18]. While some of these variations can be attributed to the design and operation of the WWTP [19], they are also likely influenced by seasonal changes. Several studies have explored the impact of seasons on MP concentrations and removal efficiencies in WWTPs, reporting considerable discrepancies in the results obtained. To elaborate, Conley et al. [20] reported no seasonal trends in MP concentration in a treatment plant in the United States, as opposed to a higher MP influx in winter in a mid-eastern WWTP reported by Ben-David et al. [21]. The lowest MPs were found during the winter season, and the highest in summer, in another study conducted in central urban Asia [22]. These contrasting findings highlight the complexity of seasonal influences on MP dynamics and underscore the need for further research to clarify these patterns across different regions, climates, and treatment systems.

Previous studies have primarily focused on MP levels in wastewater streams and their impact on the aquatic environment [22–25], whereas sludge streams remain understudied in comparison. Seasonal fluctuations of MFs are well-documented in final effluents, with fibres dominating regardless of the season [21, 26–28]. However, there are knowledge gaps in whether sludge streams follow similar trends or if the dominant MP shape, size, and polymer type shift with seasonal changes.

MPs have diverse characteristics that impact their removal through treatment processes due to differences in their size, shape, density and polymer structure. Polyethylene terephthalate (PET), polyamide (PA), polypropylene (PP), polyvinyl chloride (PVC) and polyethylene (PE) are the most frequently detected MP polymers in wastewater [29–31]. Wang et al. [32] established seasonal patterns in MP morphology and polymer profile in road dust, suggesting that stormwater, which is an important contributor to MP load

in WWTP, could potentially introduce seasonal variation of these MP parameters in sludge.

MPs also play a significant role in the environmental transport of emerging contaminants through adsorption. Studies show that a variety of pollutants are effectively sorbed by MP particles, such as heavy metals [33], antibiotics [34] and organic and inorganic compounds [35, 36], thereby contributing to secondary pollution. MPs in sludge have been found to exhibit enhanced adsorption capacity, likely due to changes in physico-chemical properties post-wastewater treatment [33]. These insights underscore the need to better understand the interactions between MPs and emerging contaminants in wastewater treatment sludges to inform effective environmental management strategies and the beneficial use of biosolids.

This study investigated the seasonal dynamics of MP concentrations and polymer profiles using sludge samples from a municipal WWTP in Ontario, a region characterized by significant temperature fluctuations between winter and summer (-35°C to 35°C) and distinct seasons. The objectives of the study were: (1) To assess the seasonal variation in MP concentration and characteristics in primary, secondary, and digested sludge streams; (2) To determine the predominant types, colours, and polymer compositions of MPs present in different sludge streams across seasons; (3) To evaluate the presence and variety of emerging contaminants adsorbed onto MPs in wastewater sludge; (4) To investigate the influence of seasonal factors, such as snowmelt and stormwater, on the abundance and types of MPs; and (5) To provide insights into the role of seasonal dynamics in the occurrence, removal and discharge of MPs from sludge treatment streams at wastewater treatment plants.

2 Materials and methods

2.1 WWTP sampling and sewershed

Samples were obtained from a WWTP in Ontario, Canada that serves a population of one million people. The facility employs an activated sludge system with nitrification for biological treatment and chemical precipitation for phosphorus removal. The city has a combined sewer system with a daily average capacity of 550 million litres per day (MLD) and an average flow of 450 MLD. The sewershed spans over $2,800\text{ km}^2$ with approximately 2,850 km of sanitary sewers and 100 km of combined sewers in the older parts of the city. Thus, approximately 3.5% of the sewer system is combined, with likely more MP pollution entering with stormwater.

For the quantification of MPs in the sludge streams, primary sludge, secondary sludge (thickened waste activated sludge), digested sludge, centrate and dewatered biosolids were collected. For the seasonal variation study, two grab samples (1–3 L each) of primary, secondary (thickened) and digested sludge were collected from the treatment plant on different days within each season – Winter (February), Spring (April), Summer (June) and Fall (September) in 2023. Samples were stored at 4°C in glass containers until the analysis was done. Digested sludge was obtained from the outlet of the anaerobic digester in the plant.

2.2 Extraction of MPs from sludge samples

MP analysis of sludges was done in duplicates with 20–50 mL subsamples. Due to the low solid content in the centrate samples, larger sample volumes of approximately 2 L

were processed. MP extraction was performed using a modified version of the method described by Rodrigues et al. [37], as outlined below. The samples were initially treated with Fenton's reagent (a 1:1 ratio of 35% H_2O_2 and 0.05 M FeSO_4). The digestion agent was added in small batches while the sample was continuously mixed. The temperature was maintained at 60 °C throughout digestion to minimize damage to the MPs. After 2 h of digestion, the samples were left for an additional 24 h after the addition of an excess of Fenton's reagent to ensure complete digestion. After 24 h, the supernatant decanted and was vacuum filtered (Filter 1) through a 10 μm nylon filter (47 mm diameter, Millipore Sigma, Darmstadt, Germany). A few drops of soap solution were added to the top layer to displace the particles towards the outlet. The soap solution was tested for MP content. The filter was washed with 35% ethanol to remove any unreacted peroxide and prevent damage to the filter itself. To the fraction after removing the supernatant, zinc chloride (ZnCl_2) solution of density 1.6–1.7 g/L was added and mixed using a magnetic stirrer for 30 min. The volume of ZnCl_2 solution added was sample-dependent to achieve a computed final solution density of approximately 1.6 g/L. This solution was left for density separation for an additional 24 h, and the supernatant was carefully filtered through a 10 μm pore size nylon filter (Filter 2). The MP sample analysis is illustrated in Fig. 1.

2.3 Quantification and physical characterisation of MPs

After extraction, suspected MPs were enumerated using microscopy with a Leica M165 C (Leica Microsystems, Germany) at magnifications of 7.3 $\times$ to 120 $\times$. The detection limit in this study is approximately 20 μm , constrained by both the capabilities of visual examination for MPs and the sensitivity of FTIR instrumentation. For systematic quantification, a custom-designed circular grid with alphanumeric coordinate markings was printed in high resolution and positioned beneath the transparent nylon filter, 47 mm in diameter. Grids provided a more precise framework, enabling tracking of segments that were counted and that remained unexamined. Given the prolonged period of counting

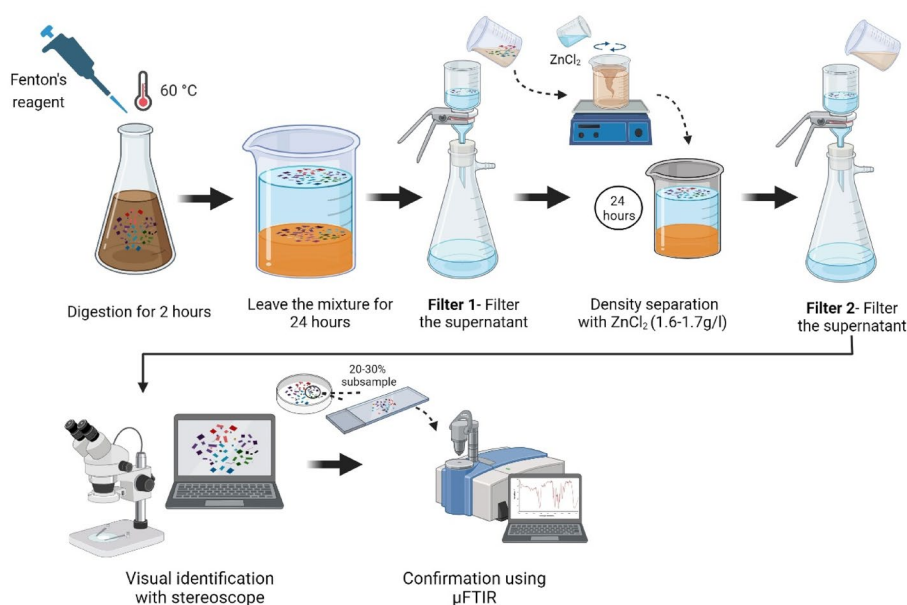


Fig. 1 Steps for MP extraction and quantification in sludge samples (Created with BioRender.com)

required for thorough analysis, this methodology was particularly important in minimizing erroneous quantification due to duplicate counting or exclusion of particles.

Although the extraction process involves both organic matter digestion and density separation, the isolated particles may still include particles other than MPs. Achieving complete elimination of organic matter is challenging, often requiring longer processing times and/or harsher protocols that can damage the integrity of MP particles while still leaving non-MP residues. Similarly, density separation is not entirely selective and can retain inorganic materials, such as straw or other low-density organic substances, alongside MPs. To address these limitations, visual sorting criteria were adapted from Hidalgo-Ruz et al. [38] and supplemented with additional guidelines based on preliminary observations. Key features included the absence of discernible organic structure, consistent thickness in fibres and the exclusion of any particles that broke down into powder or resulted in smudging on contact with the needle probe. Visual assessment alone is unreliable for separating synthetic fibres from natural (cellulose) ones, as (stereo)microscopic examination does not exhibit any distinguishable features, requiring an additional step for polymer verification with analytical instruments.

Suspected MP particles were sorted visually based on colour and shape. Primarily, two shape categories, fibres and fragments, were identified. Particles having cylindrical form with a high length to width ratio were characterized as fibres, while fragments were flat with irregular edges [39, 40]. Less common shapes were grouped under an “others” category, which comprised pellets, microbeads and foams as described by Hartmann et al. [41]. For colour classification, particles were categorized into broad colours (red, blue, black, green) with both light and darker shades grouped within the same category, along with transparent particles. Dominant colours were identified and maintained as distinct categories, whereas those observed in smaller proportions were collectively grouped under “others” in colours.

Confirmed MPs were expressed as the number of particles per gram of total solids (#MPs/gTS), where TS was determined by drying a well-mixed sample to a constant weight at 103–105 °C and calculating the ratio of dried residue to initial wet weight of the sample. The MP count was then adjusted based on the TS of the sample, which is synonymous with MPs per g dry weight used in other studies for comparison purposes.

2.4 Polymer identification and quantifying confirmed MPs

Polymer verification of a 30% subset, stratified by shape (proportional to the observed ratio of different MP types), was performed using FTIR spectroscopy (Nicolet iN10; Thermo Fisher Scientific, USA). The minimum match quality threshold was set at 60%, and spectra rounding to 60% were considered acceptable. MPs were carefully transferred to a sequentially numbered glass slide using tweezers. Very fine particles that could not be handled with tweezers were scraped onto the slide using a fine needle. The confirmed MPs were expressed as a percentage for each shape class and applied to individual filters to obtain the confirmed MP count. The OMNIC polymer spectra library (Thermo Fisher Scientific, USA) was used; if no match was found, the open-source database Open Specy [42] was consulted to identify any additional matches. Particles showing no spectral matches in both databases were excluded from the dataset to avoid false positives. When FTIR analysis yielded only compound or chemical matches, the analysis was repeated at various points on the same sample to attempt precise material identification. In cases

where only chemical or compound matches persisted, further review was conducted to evaluate if they were commonly used in plastic manufacturing or other industrial processes, and the particles were classified accordingly; additional examples are provided in Sect. 3.3.3. However, this approach was both time-consuming and impractical for every particle, as some did not withstand the ATR pressure, were lost after the first reading, or broke into fragments too small for further FTIR analysis.

2.5 Statistical analyses

Statistical analysis was conducted to assess the difference in MP characteristics (colour, type and polymer composition) across different sludge samples and seasons. Statistical difference in recovery efficiency was ascertained using one-way ANOVA. Two independent samples of each sludge type were analyzed in duplicate per season. All statistical analyses were performed at 5% significance level with MS Excel® (version 2412) using the Real Statistics Resource Pack software (Release 8.9.1) [43]. The normality of the data was verified using the Shapiro-Wilk test. Two-way ANOVA was performed to investigate the significant differences in MP characteristics among sludge types and seasons. Where applicable, Tukey's post hoc test was applied to identify which specific groups differed.

2.6 MP recovery efficiency

A recovery experiment was conducted to determine the effectiveness of the modified protocol for MP analysis. Secondary sludge was selected for the recovery experiment because it represents a more challenging matrix for MP extraction, particularly due to the presence of flocs and its higher total solids content compared to primary and digested sludge samples. MP fragments and fibres were spiked into 100 mL secondary sludge samples. The fragments (LNS Technologies, USA) were divided into three different size ranges (1.5–2.5 mm, 0.85–1.5 mm and 0.2–0.85 mm). They were imaged using Leica M165 C (Leica Microsystems, Germany) to distinguish the spiked particles from background MPs based on their documented size and morphology. Spiked fibres (sourced from everyday textile materials) measured in the 200–950 µm range with an average thickness of 25–30 µm. Table 1 shows the polymer type, densities (average values from the literature as reported by Ruggero et al. [44] and the number of MPs added. The tests were done in triplicate. Each filter required approximately 72 to 80 h for complete analysis, including the extraction of MPs, a drying period to ensure sample stability, and subsequent polymer verification to accurately identify particle composition.

2.7 Quality control steps

Several steps were taken to ensure limited contamination of MPs during sample analysis. Cotton lab coats were used during the analysis, and glass beakers were used during digestion and density separation protocols. On average, microscopic analysis of each filter took up to 8 h, and each sample consisted of two filters. Background MP

Table 1 Characteristics of spiked MPs for the recovery study

Sl no.	Sample type	Density (g/cm ³)	Colour	# particles
1.	PET fibres	1.38–1.41	White	15 each
2.	Nylon fibres	1.02–1.06	Yellow	15 each
3.	HDPE fragments	0.96	Transparent	15 each
4.	PP fragments	0.91–0.95	Green	15 each

contaminations were accounted for by enumerating the MPs in the (blank) filters left for 16 h. Long, shiny white fibres that resembled the filter material were also excluded from quantification. All water and solutions used for reagent preparation were filtered through 0.2 µm filters, minimizing contamination from preparation reagents; therefore, atmospheric deposition during sample handling and filtration was considered the primary potential source of background contamination.

3 Results and discussion

3.1 Method efficiency

The recovery experiments yielded 75–100% recovery of spiked MPs presented in Table 2, comparable to ranges reported by others [45, 46]. In the higher size range (1.5–2.5 mm), 100% of the MPs were recovered from both polymer types tested; however, efficiency dropped by over 10% with the lowest size class (0.2–0.85 mm). The reduced recovery efficiency was statistically significant for the PP polymer at the lowest size range ($p < 0.05$), whereas the HDPE polymer did not exhibit a significant reduction. Fragments had a slightly higher recovery rate compared to fibres, but the difference was not statistically significant ($p = 0.14$). There were no significant differences in the recovery of MPs among different polymer types ($p = 0.2$). The particles mostly remained intact after the extraction procedure, although some of their colours appeared lighter post-process. PET fibres had the lowest recovery at 75.5%, likely due to their white colour that blended with the nylon filter, rather than due to differences in density. Therefore, it is likely that there was a colour bias in the results, underestimating the number of white and transparent MP particles.

3.2 MP distribution in sludge streams

On average, MP concentration in sludges varied from 63 to 430 particles/gTS. The primary sludge had 221 ± 25 MPs/gTS, significantly higher ($p < 0.05$) than the other sludge samples (Fig. 2A), with an average of 2.6 times more MPs than secondary sludge (98 ± 23 MPs/gTS). Similar results are reported in other studies, where MP removal through primary treatment was found to be between 78% and 98% in different treatment plants [3, 47, 48]. Digested sludge and centrate recorded 165 ± 42 MPs/gTS and 118 ± 19 MPs/gTS, respectively. Dewatered biosolids contained 91 ± 24 MPs/g TS, significantly lower than the 228 to 1353 MPs/g TS reported by Sivarajah et al. [49], but higher than the 8–14 MPs/g TS reported by Crossman et al. [50] in Canadian biosolids. This discrepancy is likely due to differences in analytical approaches, particularly the adoption of the polymer confirmation step. Although all three studies used similar extraction protocols for organic matter removal, Sivarajah et al. [49] relied on visual assessment and physical

Table 2 Recovery rates for different MP types

SI No.	MP type	Average recovery (%)
1.	PET fibres	75.5
2.	Nylon fibres	86.7
3.	HDPE fragments (1.5–2.5 mm)	100
4.	HDPE fragments (0.85–1.5 mm)	88.9
5.	HDPE fragments (0.2–0.85 mm)	77.8
6.	PP fragments (1.5–2.5 mm)	100
7.	PP fragments (0.85–1.5 mm)	88.9
8.	PP fragments (0.2–0.85 mm)	75.6

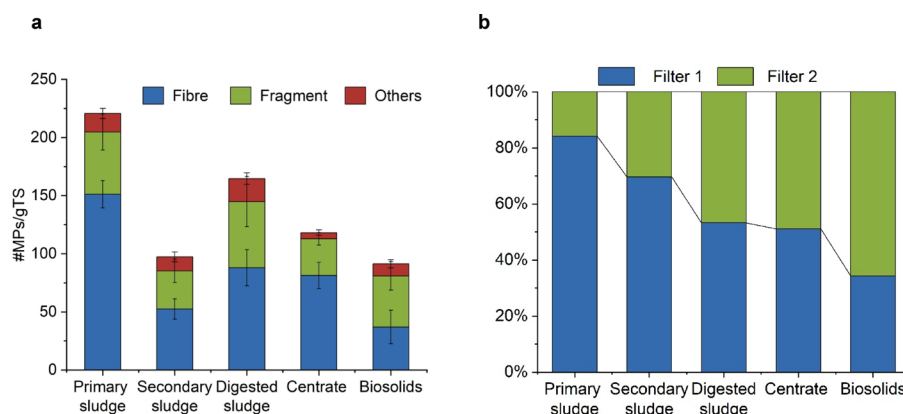


Fig. 2 **A** MP type and concentration **B** MPs distribution in filter 1 and filter 2 in different primary, secondary, digested sludges, centrate and biosolid samples

characteristics for MP classification. In contrast, Crossman et al. [50] performed polymer verification of isolated particles using FTIR spectroscopy, minimizing the inclusion of false positives. A similar approach was adopted in the present study, where an efficiency factor was calculated from the FTIR analysis of a representative subset and applied to adjust the final MP count. This further demonstrates the influence that extraction and quantification methods pose when examining similar samples, highlighting the need for a standardized protocol to enhance comparability between different studies. In addition to methodological differences, infrastructural and population size can also be influential factors. Biosolids analysed by Crossman et al. [50] were from a sewershed with a separate stormwater system and serving a smaller population. On the other hand, the sewer system in this study features a combined system in some sections of the city, serving a larger population, which likely results in higher MP influxes from both stormwater and urban sources.

Of the MP types assessed, fibres were dominantly present, followed by MP fragments and others (Fig. 2A), with the exception of biosolids, where the quantities of fibres and fragments were equivalent. The category “others” included granules, microbeads, and pellets, which were found in lower amounts. The predominant presence of fibres in all three sludge types is consistent with other studies [51–54]. MFs have been identified as a key contributor to MP pollution in domestic wastewater. Domestic laundering is the primary source of fibres in WWTPs, with an estimated MF release of 7–69 mg per kg of laundry [55]. Another significant source of MFs is atmospheric deposition, owing to their higher mobility [56–58], which eventually enters the sewers and WWTPs through stormwater runoff. MFs have been recorded as the dominant MP type in several atmospheric deposition studies resulting from the shedding of textiles due to various mechanisms such as mechanical abrasion and UV irradiation that damage the fabric [59–61]. In biosolids, no significant difference between fibres and fragments was observed ($p > 0.05$). This divergence from the fibre-dominant trend was likely due to the removal of lighter fibres during sludge dewatering at the plant, decreasing the fibre concentration in biosolids. This was further supported by the findings of Nakao et al. [62], where 75% of MFs were reported to escape into the centrate stream upon dewatering. Re-entering the wastewater stream, these MFs break down into fragments, altering the MP shape profile in dewatered sludge.

The distribution of MPs varied in the different filters (Fig. 2B). Filter 1 captures comparatively lighter particles with densities near water from the post-digestion supernatant, whereas Filter 2 isolates higher-density MPs following the ZnCl_2 density separation (Sect. 2.2). All three sludge samples had 50–80% of the MP fraction retained in Filter 1, corresponding to a significant share of comparatively lighter particles. The trend, however, reversed in the biosolids, where most particles were captured in Filter 2, as the concentrate removed lighter particles. Although larger and denser MPs are known to settle in primary clarifiers, studies have shown evidence of lower-density polymers retained at this stage [63]. The MP retention in treatment processes is more complex than the simple density-based separation used during extraction, involving other processes such as coagulation, flocculation, and flotation.

In secondary clarifiers, low-density MPs may settle due to their potential entrapment in sludge solids through extracellular polymer substances (EPS) secreted by the microbes, increasing the density of the resulting flocs [64–66]. Although the digestion process during the sample extraction can release a good amount of these MP particles, it does not ensure the complete removal of organic material. This could imply an increased density of MPs due to partial attachment to the sludge particles, resulting in a higher number of particles in Filter 2. These findings reflect the influence of sludge treatment processes, such as dewatering, on the type and density of MPs retained in the biosolids. Additionally, density separation is crucial for effectively isolating MPs from complex matrices, such as biosolids, further emphasizing the importance of extraction protocols and treatment design in MP assessment. The diversity of MP fragments and fibres isolated in this study from wastewater sludges is illustrated in Fig. 3. The figure highlights the range of colours, shapes, and polymer types present, emphasizing the heterogeneous nature of MP contamination in the waste stream.

3.3 Seasonal variation of MPs in sludges

3.3.1 Concentration and shapes of MPs

Figure 4 shows the seasonal variation of MPs in primary, secondary, and digested sludge. TS content ranged from 5.8 to 6.3% (w/w) in primary sludge, 6.9–7.5% (w/w) in thickened secondary sludge, and 2.4–2.9% (w/w) in digested sludge. Primary sludge had the highest MP levels in all four seasons, ranging from 156 to 430 MPs/g TS. MP concentrations reported in other countries were lower than those observed in the current study, with Italy and the USA recording some of the lowest levels at 1–1.7 particles per g dry

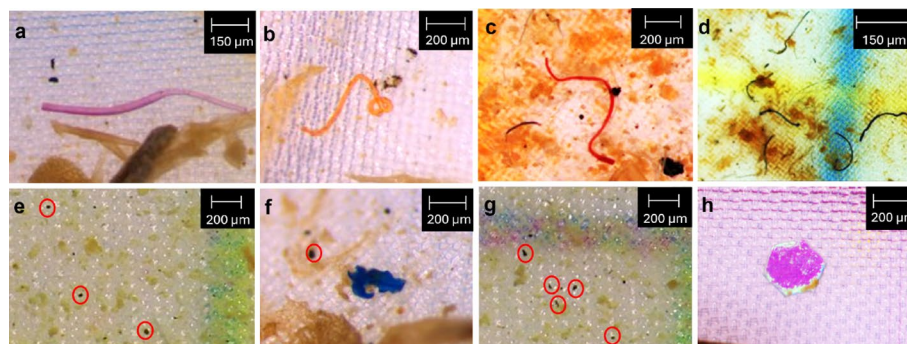


Fig. 3 MPs extracted from the wastewater sludges: Fibre (a–d) and Fragments (e–h). (a, h) from biosolids, (c, d) primary sludge, (e, f) secondary sludge, (b, g)

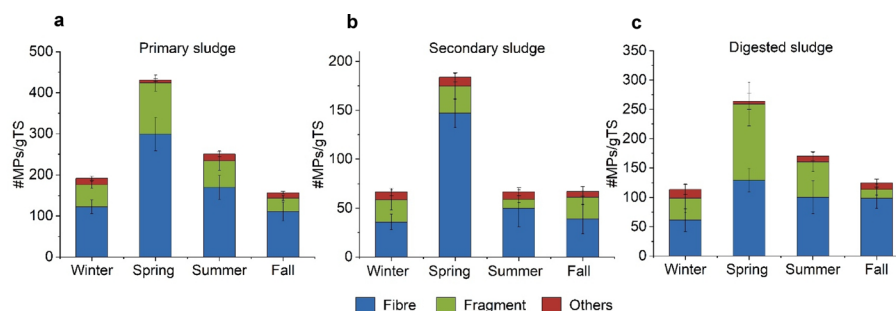


Fig. 4 Seasonal variation of MPs in **a** primary, **b** secondary and **c** digested sludge

weight [64, 67]. However, it is important to recognize the methodological factors that influence these numbers. For instance, the Italian study analyzed particles greater than 30 μm , while the US data were limited to a range of 45–400 μm . Furthermore, both studies employed different extraction protocols, which compounded the variability and limited the scope for direct comparison between the results.

Fibres remained the dominant MP shape in all the sludge samples throughout all seasons. This is consistent with the majority of studies reporting the prevalence of MFs in wastewater and sludge streams [48, 51, 68]. Although the seasonal variation in the overall composition of MP shapes (fibres vs. fragments) is insignificant ($p < 0.05$), it is worth noting that the increase in the number of fragments in spring may be linked to the tire and road wear particles (TRWPs). This is further discussed in 3.3.3.

The spring season had the highest MP concentration in all three sludge samples ($p < 0.05$). There was an average of a 1.5 to 2.5-fold increase relative to MP concentrations in other seasons. Although MP levels in primary and digested samples in summer were higher than in fall and winter, the increase was insignificant ($p > 0.05$). Winter and fall samples showed similar MP distribution in all the samples analyzed ($p > 0.05$). Similar results were observed in the WWTP study in Lithuania, with the highest MP numbers occurring in spring due to strong winds, melting ice, and heavy rainfall [69]. Furthermore, seasonal patterns are reported in atmospheric MPs, with pronounced levels in the spring season [70]. In contrast, the seasonal study by Ben-David et al. [21] reported increased MP levels in winter, likely driven by increased laundering and surface runoff typical in Middle Eastern countries. Regional climate differences may be a significant factor that can explain these discrepancies. Subzero winter temperatures are typical for Lithuania and Ontario, Canada, while temperatures in Karmiel (Israel) are consistently above 0 $^{\circ}\text{C}$ throughout the winter [71].

3.3.2 Colour analysis

Colour classification of MPs is often recommended for several reasons. Although colour cannot provide definitive source information, it can be an ancillary parameter for source tracing [72]. It is also a vital parameter in aquatic toxicology studies, where the colour of the MPs influences the ingestion preference for aquatic life [73]. The colour can also be used to assess the background contamination during sample handling. In this study, MPs were classified into broad colour categories. Black, blue, green, red and transparent were recorded colours, with a small number of other colours (Fig. 5). Black and blue colours accounted for more than 50% of the MPs recorded every season, along with a substantial number of transparent particles. The black colour was observed in comparable

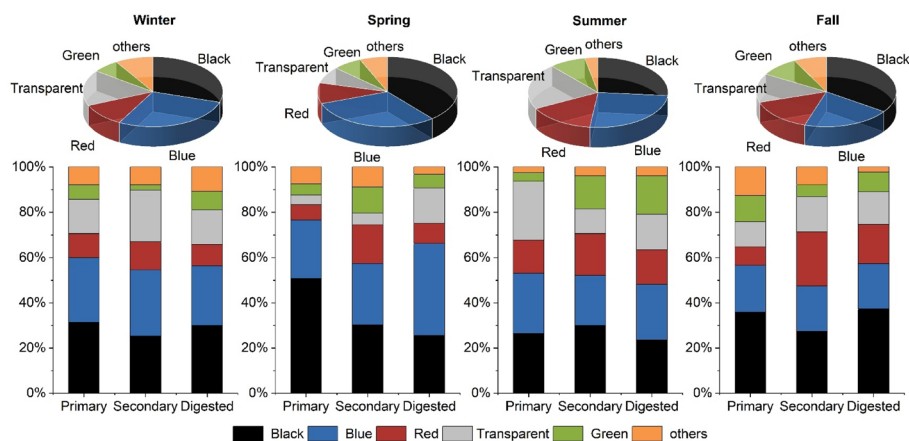


Fig. 5 MP colour distribution in primary, secondary, and digested sludge in different seasons

proportions in multiple other studies that examined MPs in WWTPs [29, 51, 74, 75], while several others reported transparent particles as the dominant type [2, 17, 76–79]. Transparent particles primarily consist of fragments and films, and they may come from different packaging materials and short-life consumer goods. Most disposable items are manufactured in white and transparent plastics [80, 81]. Although some research depicted a clear divergence and pattern among the observed colours during different seasons, Fig. 5 shows no season-dependent trend among the MP colours ($p < 0.05$).

3.3.3 Seasonal polymer composition and source attribution

The percent match to the polymers in the library varied from around 65–90%. Figure 6 shows the polymer distribution in the sludge streams in all four seasons. PET emerged as the most detected polymer type across all seasons, with other polymers, such as PA, PP, and PE (including LDPE and HDPE), also present in substantial amounts. PET concentrations fluctuated across different sludge types and seasons. The primary sludge had the highest fraction of PET (39.5%, combining all the seasons), with seasonal variations ranging from 32.5–46.4%. Secondary sludge exhibited lower amounts of PET, with an average of 31.5% (27.4–34.8%), while anaerobically digested sludge contained 27.9% PET. A statistically insignificant deviation was observed in the digested sludge samples in summer, where PET and PE were present in similar fractions but at higher concentrations than PA. Similar to this study, the abundance of PET polymer was also reported among isolated MPs in WWTPs in different countries [19, 30, 82, 83]. An atmospheric deposition study in Southern Ontario by Welsh et al. [84] further complements the current trend in polymer concentrations, revealing significant levels of PET and PA. While both polymers are known to be important components in the textile and apparel industry [85–87], PET has more versatile applications in packaging, including plastic bottles [88]. Research also shows that, on average, polyester/PET-infused clothing sheds over five times more than nylon apparel. PET shedding increased tenfold in the fleece-type fabrics [89] that are increasingly used in colder urban climates. These factors likely explain the prevalence of PET in wastewater sludges.

Some particles were matched to compounds that may be used in the manufacture of plastics and additives, including plasticizers and flame retardants, used in plastic materials. Pentafulpropionic acid, Tegin 90, phosphorus oxychloride and ethylbenzene are

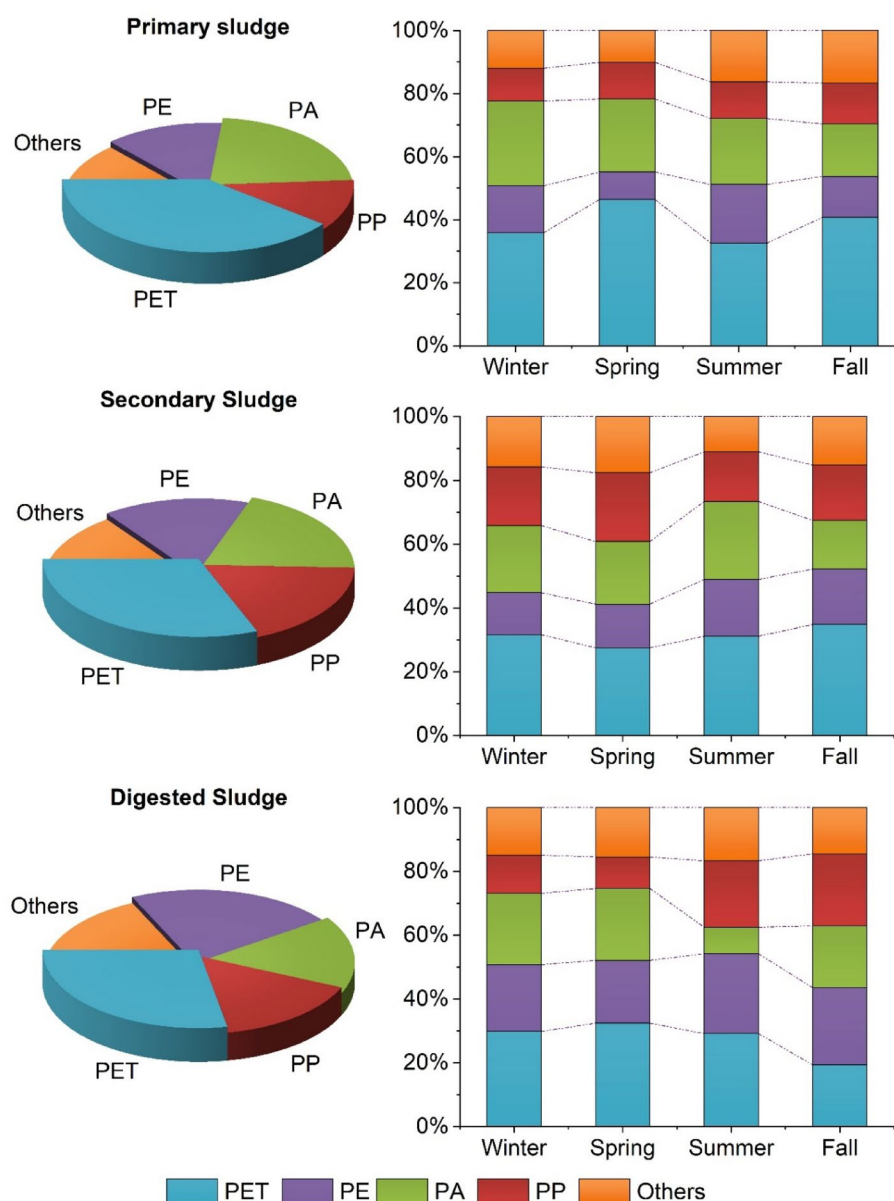


Fig. 6 Analysis of polymer type distribution in sludge samples across different seasons

examples of such compounds. In some cases, due to weathering or aging of the material, the match to the polymer itself proved difficult; instead, it resulted in a match to the additives used in plastic manufacturing. These matches do not directly provide information about the polymer itself, but they do reveal direct links to the materials that suggest a plastic nature and are grouped under the category “others.” This is illustrated further in Fig. 7. Apart from these, trace amounts of other polymers and their derivatives were also recorded in this class.

Rubber, additives used in tires and road marking, are classified under “tire and rubber-related compounds” (Fig. 7). The spring season showed a relatively higher proportion of particles classified under this category. Several factors may contribute to this observation. Switching to winter tires is a standard safety practice in cold countries, and over 65% of Ontario drivers, on average, make this switch each winter [90]. Winter tires

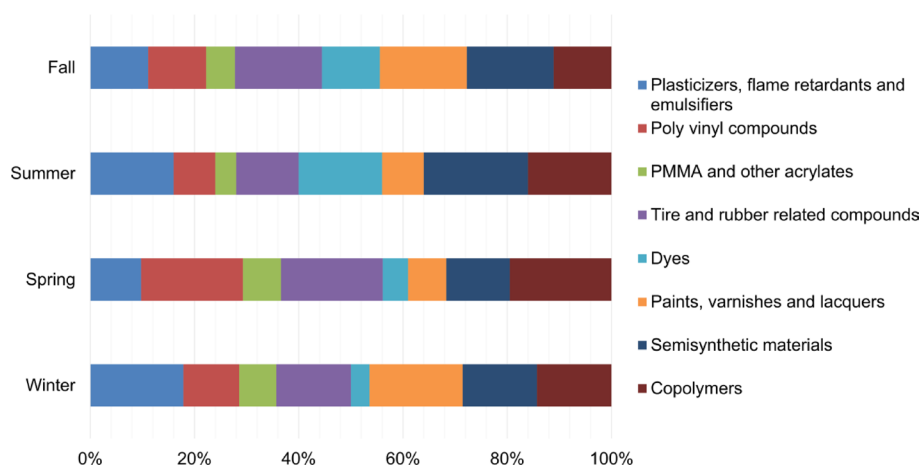


Fig. 7 Detailed breakdown of the category 'others' in seasonal polymer distribution in sludge

produce substantially more wear particles, approximately 10 times more than non-stud-
ded summer tires [91], which could contribute to increased MP content in spring due
to snowmelt and stormwater. High levels of road dust were recorded in late winter and
early spring in Sweden [92], whereas a Chinese study reported no significant seasonal
variation in road dust abundance [32]. These contrasting results suggest that snowfall
and temperature likely play a key role. Ma'anshan city experiences mild winters, whereas
Stockholm endures comparatively harsher winter conditions that may have contrib-
uted to the increased release of road dust during thaw periods in the above-mentioned
studies.

Yet another possible source of MPs can be the road de-icing salts. A study by Rød-
land et al. [93] found that road de-icing salts contained 32–252 MPs/kg, where an
overwhelming 96% were reported to be black rubber-like particles. These findings are
particularly important in Canada, where approximately 5 million tonnes of road salt are
used annually, with urban centers leading in per-area usage [94]. For instance, Ontario's
winter road maintenance requires 1.5 million tonnes of road salt, with urban areas like
Ottawa using 185,000 tonnes of road salt each winter to maintain road safety [95, 96]. In
countries with prolonged winters and heavy snow loads, de-icing may be a critical MP
contributor to WWTPs. These factors, combined with the restricted movement of MPs
due to the presence of snow and ice in winter, can explain the high contribution of MPs
in the spring as the snow and ice melt.

The dominance of fibres and PET polymer across all the seasons along with the con-
sistent prevalence of black and blue colored MPs, indicates that seasonal variation has
limited influence on MP characteristics. These patterns further suggest that source
contributions remained relatively stable throughout the study period. In contrast, sea-
sonal differences were more prominent in MP abundance, likely reflecting the impact
of hydrological conditions and runoff dynamics into the wastewater streams. Therefore,
seasonality appears to primarily affect MP levels than their composition.

3.3.4 Analysis of particles excluded from MP classification

Particles that did not meet the criteria for classification as MPs through FTIR analysis
were examined in detail further (Fig. 8). More than 40% of the discarded particles were
natural fibres, mainly cellulose. They were not removed during the digestion and density

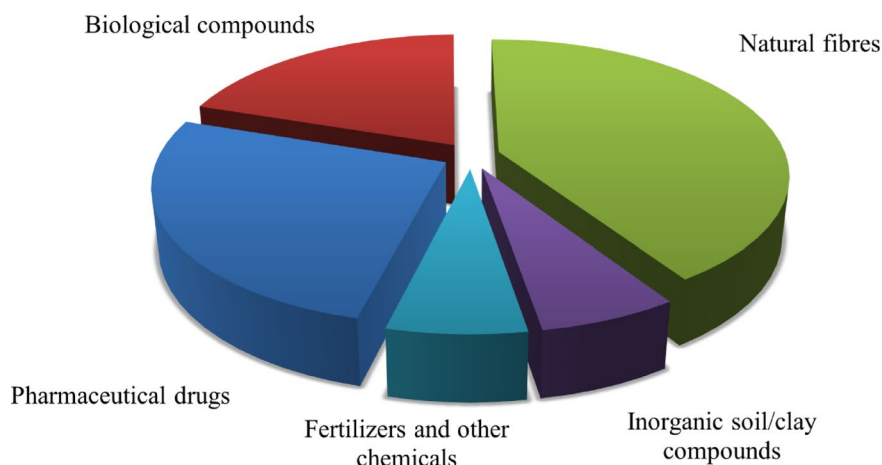


Fig. 8 Excluded compounds in suspected MPs: Distribution overview

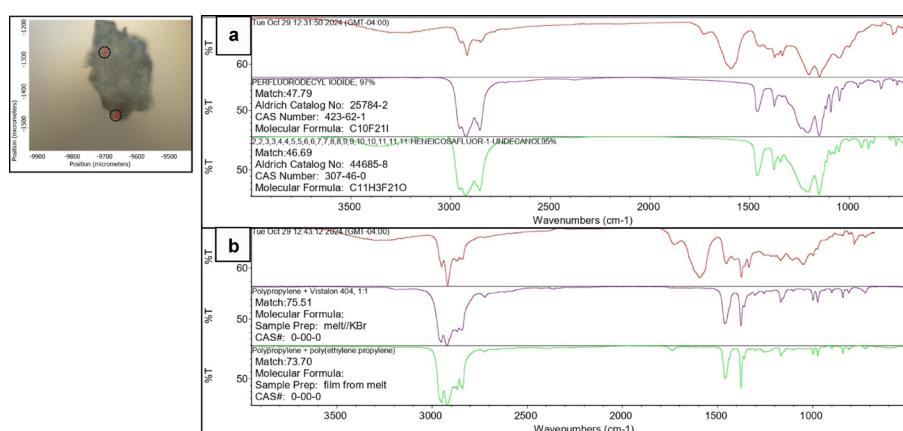


Fig. 9 **A** Spectra of PFAS compound adsorbed on suspected MP **B** Polymer nature of the MP fragment

separation stages. Cellulose is often tagged as a leading interferent in MP analysis of wastewater samples [17, 97]. Multiple digestion steps are necessary to effectively break down and remove cellulose [98]. This prolongs the analysis period and can also damage the MP particles in the samples. Because cellulose fibres did not look any different from the MP fibres under the microscope, they were misidentified as of plastic origin during microscopy.

Besides cellulose, the polymer nature of other particles was unknown. 26% of the suspected particles matched pharmaceutical drugs and their intermediates, such as ibuprofen, pyrazole, methamphetamines, and 2-pyridinecarboxaldehyde. Several suspected MP particles had spectral signatures corresponding to fertilizers and other industrial chemical compounds. While these compounds could have originated from the parent particles, many were found on the fibres. This indicates a higher chance that these chemicals may have been adsorbed onto particles, some of which may be of synthetic origin. In some instances, multiple compounds were detected on a single particle. For example (Fig. 9A), at sample point 1, the match was to perfluorodecyl iodide (97%), which is a PFAS compound, while at sample point 2 (Fig. 9B), the match was found to be for PP. This demonstrates the adsorption behaviour of MPs and the presence of emerging contaminants adsorbed on MPs. The adsorption capacity of MP particles is primarily

associated with their large specific surface area and hydrophobicity [99]. Strong evidence that the MPs act as vectors for other emerging contaminants, such as fertilizer chemicals and pharmaceuticals, is well supported by recent studies [34, 100]. MPs may thus accumulate other pollutants, exacerbating their environmental impact and toxicity on aquatic and terrestrial ecosystems.

4 Limitations of the study and future research

Although ongoing efforts are improving the standardization of MP detection and quantification, methodological differences remain important when reporting MP concentrations and comparing results among studies in different matrices. The present study provides a detailed assessment of seasonal MP abundance and characteristics across sludge streams from a municipal WWTP, while also identifying methodological considerations and limitations that can guide future investigations.

The analytical detection limit of approximately 20 μm focused the analysis on particles reliably detectable by the microscopy and FTIR analysis. Consequently, smaller MPs and nanoplastics were not evaluated. Future studies could extend the size range using higher-resolution techniques, such as laser direct infrared (LDIR) spectroscopy. The reported MP concentrations in this study were not corrected for recovery efficiency, as the recovery experiment was conducted to provide a general estimate of method performance and showed variation with particle size, polymer type, and shape; therefore, applying a single correction factor was not considered appropriate. Instead, the recovery results provide context for interpreting the reported concentrations. It is recommended that future studies develop and adopt standardized recovery-test protocols to enable more consistent evaluation of method performance and improve comparability across MP studies.

Additionally, this study used FTIR data to indicate variation in the types and amounts of contaminants potentially adsorbed onto MPs. More targeted, high-resolution chemical analysis would help confirm and quantify these contaminants using advanced techniques such as GC-MS/MS and Py-GC-MS. Future research should extend seasonal monitoring to different contributing sources including stormwater systems to better characterize MP pathways and their contribution to sludge. Studies across WWTPs with different treatment configurations, sewer systems and characteristics would further test the applicability of the seasonal patterns observed here. Higher-frequency sampling during rainfall and snowmelt events could improve source attribution, particularly for tire and road-wear particles.

5 Conclusion

This study explored the influence of seasons on the type and concentration of MPs in primary, secondary and digested sludge samples. Primary sludge had the highest MP levels across all seasons. Peak MP levels occurred in the spring, along with higher levels of tire and rubber-related particles. The higher abundance of MPs was likely due to run-off from snowmelt during spring, following prolonged subzero temperature conditions during winter. The most prevalent polymer type was PET in both primary and secondary sludge, while the fibres were the most dominant MP type. PA, PP, and PE (both high- and low-density) were also found in substantial amounts in various sludge samples. This remained consistent across all seasons and different sludge samples. MP colour

distribution revealed that blue and black were more commonly found colours, and their distribution did not vary seasonally. These results point out that seasonal variation influenced MP concentrations rather than their characteristics such as shape, polymer type and color distribution. Several other contaminants, including fertilizers and pharmaceutical drugs, were found adsorbed onto suspected MPs, often obscuring their underlying material properties and potentially leading to an underestimation of MP abundance in sludge samples. This finding also supports the role of MPs as carriers of emerging contaminants, such as PFAS. The results further demonstrate that MP levels in sludge fluctuate seasonally, which can directly influence their concentrations in soils receiving biosolid applications and in surface waters after wastewater effluent discharges, particularly in the spring. Understanding these fluctuations enables the development of more targeted intervention strategies that are responsive to periods of heightened risk, thereby helping to mitigate the environmental impacts of MPs.

Abbreviations

FTIR	Fourier transform infrared
MF	Microfibres
MP/s	Microplastics
PA	Polyamide
PE	Polyethylene
PET	Polyethylene terephthalate
PP	Polypropylene
TRWPs	Tire and road wear particles
WWTP	Wastewater treatment plant
ZnCl ₂	Zinc chloride

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1007/s43832-026-00440-0>.

Supplementary Material 1.

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Author contributions

N.C.: Conceptualization, Investigation, Methodology, Data Curation, Formal analysis, Visualization, Writing—original draft; J.V.: Conceptualization, Writing—Review & Editing, Resources, Funding acquisition; S.A.S.: Conceptualization, Writing—Review & Editing, Resources, Funding acquisition; J.P.: Conceptualization, Writing—Review & Editing, Resources, Funding acquisition; B.Ö.: Conceptualization, Methodology, Writing—Review & Editing, Resources, Funding acquisition, Supervision, Validation.

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Data availability

Data supporting the findings of this study are provided within the manuscript. The Supplementary Information also includes representative FTIR spectra from an indicative subset of samples. Indicative raw FTIR data are available in Figshare at <https://doi.org/10.6084/m9.figshare.32180061>.

Declarations

Ethics approval and consent for participate

This is not applicable.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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